

Savitribai Phule Pune University
S.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination
Lab Course CS-271-VSC-P
Advanced Python Programmin

Duration: 3 Hours

Maximum Marks:35

Q1. Write Python code to create a dictionary containing mobile names and their prices. Create DataFrame from this dictionary. Display the contents of DataFrame.

[10 Marks]

Answer

```
import pandas as pd
```

```
# Step 1: Create a dictionary with mobile names and prices
```

```
mobile_data = {  
    "Mobile": ["iPhone 15", "Samsung Galaxy S23", "OnePlus 11", "Redmi Note 12",  
    "Realme Narzo 60"],  
    "Price": [79999, 74999, 56999, 17999, 19999]  
}
```

```
# Step 2: Create a DataFrame from the dictionary
```

```
df = pd.DataFrame(mobile_data)
```

```
# Step 3: Display the contents of the DataFrame
```

```
print("Mobile DataFrame:")  
print(df)
```

Q2. Write a python program to list all files in a current directory. Display total count of .py files and content of any one .py files

[20 Marks]

Answer

```
import os
```

```
# Get current directory
```

```
path = os.getcwd()
```

```
# List all files in the directory
```

```
files = os.listdir(path)
```

```
print("Files in current directory:")  
for file in files:
```

```
print(file)

# Filter .py files
py_files = [f for f in files if f.endswith(".py")]

# Display total count of .py files
print("\nTotal number of .py files:", len(py_files))

# Display content of one .py file
if py_files:
    file_name = py_files[0] # taking the first .py file
    print("\nDisplaying content of:", file_name)

    with open(file_name, "r") as f:
        content = f.read()
        print(content)
else:
    print("\nNo .py files found in the directory.")
```

Q3.Viva

[5 Marks]

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Q1. Write a python program to accept a string, Count total number of vowels, consonants and special symbols from it.

[10 Marks]

ANSWER

```
# Accept a string from user
s = input("Enter a string: ")

vowels = 0
consonants = 0
special = 0

for ch in s:
    if ch.isalpha(): # check if character is alphabet
        if ch.lower() in "aeiou":
            vowels += 1
        else:
            consonants += 1
    else:
        special += 1 # count digits, spaces, and symbols as special

print("Total Vowels:", vowels)
print("Total Consonants:", consonants)
print("Total Special Symbols:", special)
```

Q2. Write a Python program to demonstrate a user-defined exception that checks whether the entered marks are between 0 and 100. Display an appropriate message for invalid input and out-of-range values. Also display Grade of marks.

[20 Marks]

ANSWER

```
# User-defined exception
class InvalidMarksError(Exception):
    pass
```

```
try:
    marks = int(input("Enter marks (0-100): "))

    # Check range
    if marks < 0 or marks > 100:
        raise InvalidMarksError("Marks should be between 0 and 100")

    # Display grade
    if marks >= 90:
        grade = "A"
    elif marks >= 75:
        grade = "B"
    elif marks >= 60:
        grade = "C"
    elif marks >= 40:
        grade = "D"
    else:
        grade = "F"

    print("Grade:", grade)

except ValueError:
    print("Invalid input! Please enter numeric marks.")

except InvalidMarksError as e:
    print("Error:", e)
```

Q3. Viva

[5 Marks]

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Q1. Write a Python program that accepts value of m and n. Divides m by n and handles a ZeroDivisionError if the user tries to divide by zero.

[10 Marks]

ANSWER

try:

```
m = int(input("Enter value of m: "))  
n = int(input("Enter value of n: "))
```

```
result = m / n  
print("Result =", result)
```

except ZeroDivisionError:

```
print("Error: Division by zero is not allowed.")
```

except ValueError:

```
print("Error: Please enter valid numeric values.")
```

Q2. Write a python program to accept a file name from user. If file exist display contents of file and number of characters, words and lines in a file, otherwise display appropriate message

[20 Marks]

ANSWER

import os

Accept file name from user

```
fname = input("Enter file name: ")
```

Check if file exists

if os.path.exists(fname):

```
    with open(fname, "r") as f:  
        content = f.read()
```

```
    print("\nFile Contents:\n")  
    print(content)
```

```
# Count characters, words, and lines
char_count = len(content)
word_count = len(content.split())
line_count = len(content.splitlines())

print("\nNumber of Characters:", char_count)
print("Number of Words:", word_count)
print("Number of Lines:", line_count)

else:
    print("File does not exist.")
```

Q3. Viva

[5 Marks]

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Q1. Write a Python program to create a list of elements. Shuffle the elements of a given list and display the list. [Hint: Use random.shuffle()]

[10 Marks]

ANSWER

```
import random

# Create a list
my_list = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

print("Original List:")
print(my_list)

# Shuffle the list
random.shuffle(my_list)

print("\nShuffled List:")
print(my_list)
```

Q2. Write a python program for following

- a) Create a file which stores the first name, middle name and last name of your 5 friends
- b) Display contents of file
- c) Append other 5 friend's names
- d) Display modified file contents

[20 Marks]

ANSWER

```
# a) Create a file with 5 friends' names
friends = [
    "John Michael Doe\n",
    "Emma Rose Smith\n",
    "Liam Alexander Brown\n",
    "Olivia Grace Johnson\n",
    "Noah William Davis\n"
]
```

```

with open("friends.txt", "w") as f:
    f.writelines(friends)

print("File created with 5 friends' names.\n")

# b) Display contents of file
print("Contents of file:")
with open("friends.txt", "r") as f:
    content = f.read()
    print(content)

# c) Append other 5 friends' names
more_friends = [
    "Sophia Marie Wilson\n",
    "Mason James Moore\n",
    "Ava Elizabeth Taylor\n",
    "Ethan Daniel Anderson\n",
    "Isabella Lily Thomas\n"
]

with open("friends.txt", "a") as f:
    f.writelines(more_friends)

# d) Display modified file contents
print("Modified file contents:")
with open("friends.txt", "r") as f:
    content = f.read()
    print(content)

```

Q3. Viva

[5 Marks]

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Q1. Write a python program to accept a string, find all words that do not contain vowels

[10 Marks]

ANSWER

```
# Accept string from user
text = input("Enter a string: ")

# Define vowels
vowels = "aeiouAEIOU"

# Split the text into words
words = text.split()

# Find words without vowels
words_no_vowels = [word for word in words if not any(ch in vowels for ch in word)]

print("Words without vowels:", words_no_vowels)
```

Q2. Write a python program to create a list of prime numbers between m and n using List comprehension. (Accept the value of m and n from user, m should be less than n)

[20 Marks]

ANSWER

```
# Accept values from user
m = int(input("Enter value of m (lower bound): "))
n = int(input("Enter value of n (upper bound): "))

# Ensure m < n
if m >= n:
    print("Error: m should be less than n.")
else:
    # List comprehension to find primes
    primes = [num for num in range(m, n+1)
```

```
if num > 1 and all(num % i != 0 for i in range(2, int(num**0.5)+1))]  
print(f"Prime numbers between {m} and {n}:", primes)
```

Q3. Viva

[5 Marks]

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Q1. Write a Python program that prompts the user to input two numbers and raises a TypeError exception if the inputs are not numerical.

[10 Marks]

ANSWER

try:

Accept inputs from user

num1 = input("Enter first number: ")

num2 = input("Enter second number: ")

Check if inputs are numeric

if not (num1.replace('.', '', 1).isdigit() and num2.replace('.', '', 1).isdigit()):
raise TypeError("Both inputs must be numerical.")

Convert to float

num1 = float(num1)

num2 = float(num2)

print("You entered numbers:", num1, "and", num2)

except TypeError as e:

print("TypeError:", e)

Q2. Create a following DataFrame named as “data”. Write the python code for the following

	Company	Count	Price
pencil	Apsara	15	250
pencil	Nataraj	20	200
pen	cello	25	600
pen	Pakar	35	900
eraser	Apsara	20	300

a) Find all rows with the label “Pencil”. Extract all columns

- b) Change the count of Eraser as 25 instead of 20.
- c) List only the columns Company and Price.
- d) List only rows with labels 'Pencil' and 'Pen'
- e) Delete column Count from the above DataFrame and display DataFrame.

[20 Marks]

ANSWER

```
import pandas as pd

# Create DataFrame
data = pd.DataFrame({
    "Company": ["Pencil", "Pencil", "Pen", "Pen", "Eraser"],
    "Name": ["Apsara", "Natraj", "Cello", "Parkar", "Apsara"],
    "Count": [15, 20, 25, 35, 20],
    "Price": [250, 200, 600, 900, 300]
})

print("Original DataFrame:\n", data)

# a) Find all rows with the label "Pencil" (Company == "Pencil")
pencil_rows = data[data["Company"] == "Pencil"]
print("\nRows with Company 'Pencil':\n", pencil_rows)

# b) Change the count of Eraser as 25 instead of 20
data.loc[data["Company"] == "Eraser", "Count"] = 25
print("\nDataFrame after changing Eraser count:\n", data)

# c) List only the columns Company and Price
company_price = data[["Company", "Price"]]
print("\nColumns Company and Price:\n", company_price)

# d) List only rows with labels 'Pencil' and 'Pen'
pencil_pen_rows = data[data["Company"].isin(["Pencil", "Pen"])]
print("\nRows with Company 'Pencil' or 'Pen':\n", pencil_pen_rows)

# e) Delete column Count from the DataFrame and display
data.drop(columns=["Count"], inplace=True)
print("\nDataFrame after deleting 'Count' column:\n", data)
```

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[5 Marks]

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Q1. Write a python program to accepts value of radius from user and calculate the area of a circle using function. (Use 3.14 as the value of π).

[10 Marks]

ANSWER

```
# Function to calculate area of a circle
def circle_area(radius):
    pi = 3.14
    return pi * radius * radius

# Accept radius from user
try:
    r = float(input("Enter the radius of the circle: "))
    if r < 0:
        print("Radius cannot be negative.")
    else:
        area = circle_area(r)
        print(f"Area of the circle with radius {r} is: {area}")
except ValueError:
    print("Invalid input! Please enter a numeric value for radius.")
```

Q2. Write a python program to create a list, string and tuple in python. Apply random module to display unique multiple random elements from above created sequence type.

[10 Marks]

ANSWER

```
import random

# Create sequences
my_list = [1, 2, 3, 4, 5, 6, 7, 8, 9]
my_string = "PythonProgramming"
my_tuple = ('a', 'b', 'c', 'd', 'e', 'f')

# Number of elements to pick (less than or equal to length)
num_list = 4
```

```
num_string = 5
num_tuple = 3
```

```
# Random unique selections
```

```
random_list = random.sample(my_list, num_list)
random_string = random.sample(my_string, num_string)
random_tuple = random.sample(my_tuple, num_tuple)
```

```
print("Original List:", my_list)
print("Random unique elements from List:", random_list)
```

```
print("\nOriginal String:", my_string)
print("Random unique characters from String:", random_string)
```

```
print("\nOriginal Tuple:", my_tuple)
print("Random unique elements from Tuple:", random_tuple)
```

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[5 Marks]

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Q1. Write a python program to accepts a value of n and create a generator that yields even numbers up to n.

[10 Marks]

ANSWER

```
# Function to create a generator for even numbers
def even_numbers(n):
    for i in range(2, n+1, 2): # Start from 2 and step by 2
        yield i

# Accept value of n from user
try:
    n = int(input("Enter a value for n: "))
    if n < 2:
        print("No even numbers up to", n)
    else:
        print(f"Even numbers up to {n}:")
        for even in even_numbers(n):
            print(even, end=" ")
except ValueError:
    print("Invalid input! Please enter an integer.")
```

Q2. Write a python program for following -

- a) Create DataFrames Student 1, Student 2 and display contents
- b) Join the two DataFrames in third frame with matching records from both sides.
- c) Display Third DataFrame

[20 Marks]

Student 1

	ID	NAME	AGE
0	S1	Anita	18
1	S2	Sakshi	23
2	S3	Om	20

3	S4	Raj	21
4	S5	Kanika	20

Student 2

	ID	City	Marks
0	S2	Pune	97
1	S4	Nashik	78
2	S5	Mumbai	86
3	S6	Delhi	66
4	S7	Nandel	54

ANSWER

import pandas as pd

a) Create DataFrames Student 1 and Student 2

```
student1 = pd.DataFrame({
    "ID": ["S1", "S2", "S3", "S4", "S5"],
    "Name": ["Anita", "Sakshi", "Om", "Raj", "Kanika"],
    "Age": [18, 23, 20, 21, 20]
})
```

```
student2 = pd.DataFrame({
    "ID": ["S2", "S3", "S5", "S6"],
    "Name": ["Sakshi", "Om", "Kanika", "Rahul"],
    "Age": [23, 20, 20, 22]
})
```

```
print("Student 1 DataFrame:\n", student1)
print("\nStudent 2 DataFrame:\n", student2)
```

b) Join the two DataFrames on ID and Name (inner join to get matching records)
third_df = pd.merge(student1, student2, on=["ID", "Name", "Age"], how="inner")

c) Display Third DataFrame
print("\nThird DataFrame with matching records:\n", third_df)

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[5 Marks]

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Q1. Write a python program to create a DataFrame from list of 10 integer values. Display contents of DataFrame and calculate sum, mean, median, mode, Standard deviation of DataFrame.

[10 Marks]

ANSWER

```
import pandas as pd
from scipy import stats # For mode calculation

# List of 10 integers
numbers = [12, 25, 7, 9, 25, 18, 30, 7, 15, 20]

# Create DataFrame
df = pd.DataFrame(numbers, columns=["Numbers"])

# Display DataFrame
print("DataFrame:\n", df)

# Calculate statistics
sum_value = df["Numbers"].sum()
mean_value = df["Numbers"].mean()
median_value = df["Numbers"].median()
mode_value = df["Numbers"].mode().values # Returns array of modes
std_value = df["Numbers"].std()

print("\nStatistics:")
print("Sum:", sum_value)
print("Mean:", mean_value)
print("Median:", median_value)
print("Mode:", mode_value)
print("Standard Deviation:", std_value)
```

Q2. Write a Python program to create a class that acts as an iterator and generates the factorial series up to n ($1! + 2! + 3! + \dots + n!$). (Accept input from the user)

[20 Marks]

ANSWER

```

# Import factorial function
from math import factorial

# Class to generate factorial series
class FactorialSeries:
    def __init__(self, n):
        self.n = n
        self.current = 1 # Start from 1

    def __iter__(self):
        return self

    def __next__(self):
        if self.current > self.n:
            raise StopIteration
        else:
            fact = factorial(self.current)
            self.current += 1
            return fact

# Accept input from user
try:
    n = int(input("Enter a positive integer n: "))
    if n <= 0:
        print("Please enter a positive integer greater than 0.")
    else:
        print(f"Factorial series up to {n}:")
        series = FactorialSeries(n)
        total_sum = 0
        for value in series:
            print(value, end=" ")
            total_sum += value
        print(f"\nSum of factorials: {total_sum}")

except ValueError:
    print("Invalid input! Please enter an integer.")

```

Q3. Viva

[5 Marks]

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Q1. Write a python code to a create a list of 10 integer numbers. Display square of all elements of list.(Use a lambda function with map())

[10 Marks]

ANSWER

```
# Create a list of 10 integers
numbers = [2, 5, 7, 3, 9, 4, 6, 8, 1, 10]

# Use lambda and map to calculate squares
squares = list(map(lambda x: x**2, numbers))

# Display results
print("Original List:", numbers)
print("Squares of elements:", squares)
```

Q2. Write a Python program that accepts a string from the user and performs the following tasks:

- a) Split the string into individual words.
- b) Identify and display all words that do not contain any vowels (a, e, i, o, u), ignore case while checking for vowels.
- c) Sort and display these words in alphabetical order.
- d) Display the total count of such words.

[20 Marks]

ANSWER

```
# Accept string input from user
text = input("Enter a string: ")

# a) Split the string into individual words
words = text.split()

# b) Identify words without vowels (case-insensitive)
no_vowel_words = [word for word in words if not any(ch.lower() in "aeiou" for ch in word)]
```

c) Sort the words alphabetically

no_vowel_words_sorted = sorted(no_vowel_words, key=str.lower)

d) Display results

print("\nWords without vowels:", no_vowel_words_sorted)

print("Total count of words without vowels:", len(no_vowel_words_sorted))

Q3. Viva

[5 Marks]